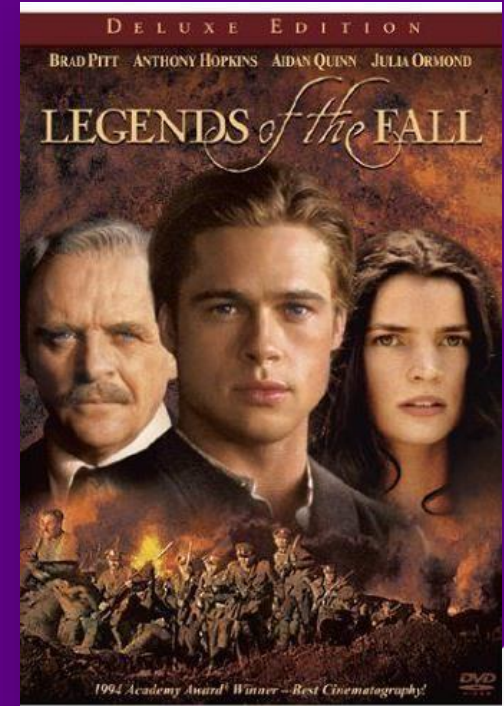


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Introduction: Welcome to the world of movie recommendations

The rapid development of data collection has heralded a new era of information. Data is being used to create more efficient systems, which is where Recommendation Systems come into play **What is a recommender system?**

A recommender system serves as a friendly guide who understands your tastes and makes appropriate recommendations. Rather than depending on subjective opinions, these systems use data and algorithms to assess your prior decisions and actions and make targeted recommendations. They estimate what you'll like next based on criteria such as your previous interactions, ratings, and even resemblance to other users. Their objective is to improve your experience by recommending material that matches your interests.

The two most common recommendation system techniques are:

- 1) Collaborative filtering
- 2) Content-based filtering

Exploring Feedback Data

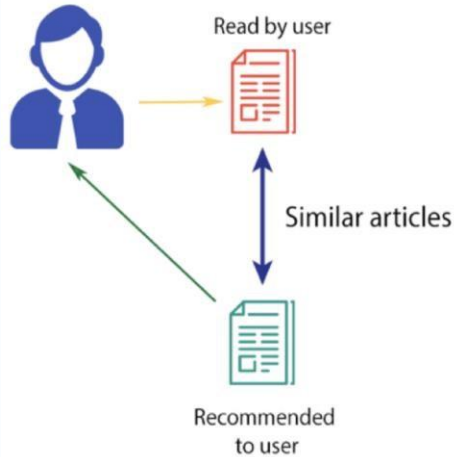
There are two kinds of feedback data:

Explicit feedback considers a user's direct response to an item (e.g., rating, like/dislike).

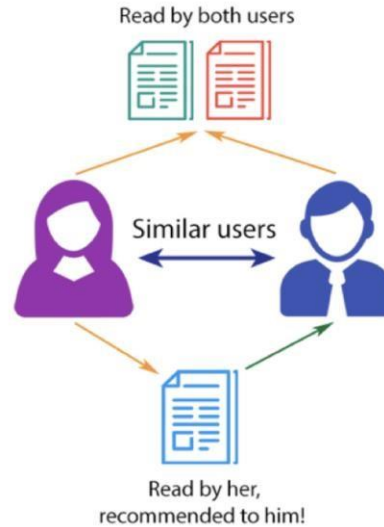
Implicit feedback examines a user's indirect behavior toward an object (e.g., the number of times a user has watched a movie).



CONTENT-BASED FILTERING



COLLABORATIVE FILTERING



The Recommendation Techniques

The 2 major techniques are :

- 1) Collaborative filtering
- 2) Content-based filtering

Collaborative Filtering : Similar users like similar things. It uses item preferences from other users to predict which item a particular user will like best. It can be further divided into two categories- memory-based and model-based.

Content-based Filtering : User and item features. Given a set of item features (movie genre, country). It predicts how a

user will rate an item based on their ratings of previous movies.

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- Stream254 understands that traditional one-size-fits-all approaches to content delivery are no longer adequate to

Business Problem

suit the changing needs of Kenyan consumers.

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- Users who do not receive tailored recommendations may experience choice fatigue and dissatisfaction while traversing the platform's enormous content library.
- In a highly competitive market, failing to engage users effectively may result in lower retention rates and a loss of revenue share.
- To address this issue, Stream254 intends to develop a recommendation system to assess user data and provide highly relevant movie recommendations.

- To also improve customer satisfaction, drive long -term engagement, and establish itself as a leader in the Kenyan streaming industry.

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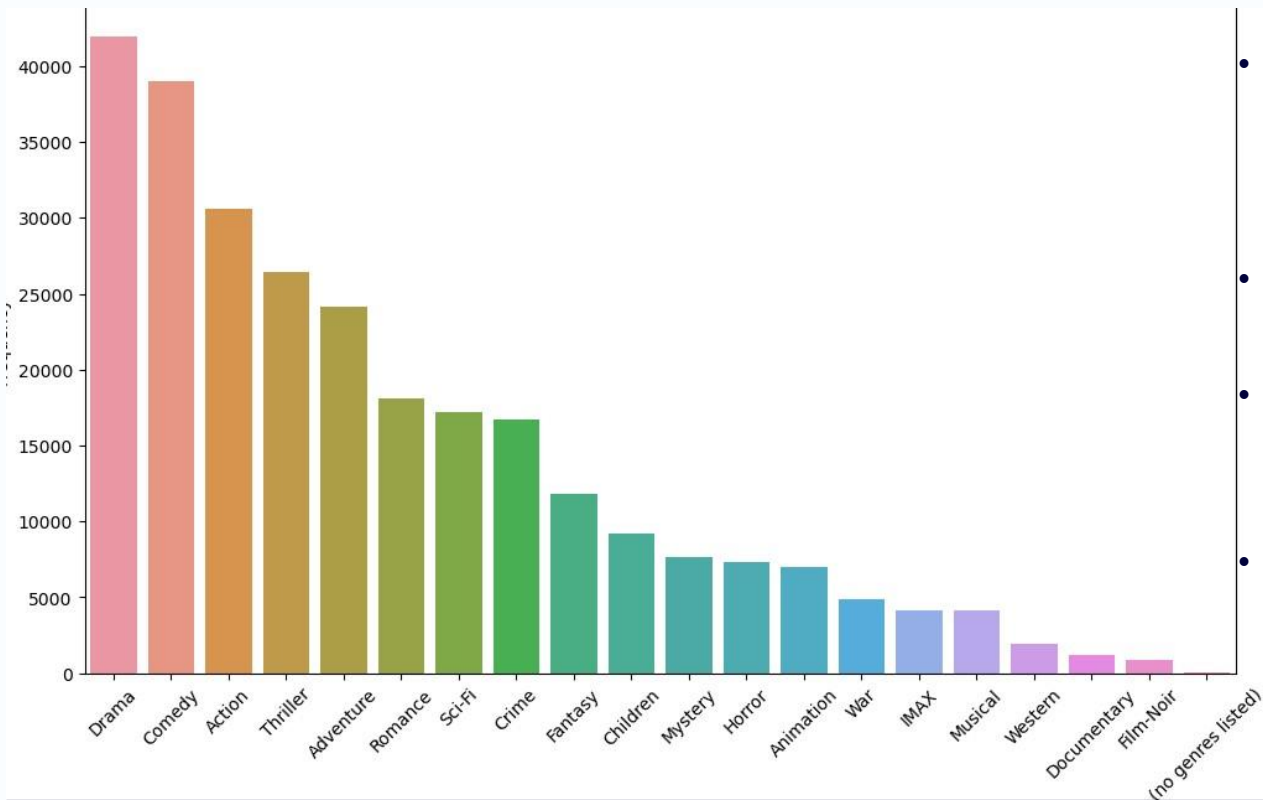
Business Objective

Develop and implement a recommendation algorithm that will analyze user behaviour, preferences, and viewing patterns to generate personalized movie recommendations with the intention to:

- Enhance User Satisfaction: By providing users with tailored movie recommendations, we aim to enhance user satisfaction and deliver an immersive viewing experience that keeps users coming back for more.

- **Increase User Retention:** Through personalized recommendations, we seek to increase user retention rates and encourage users to spend more time on the platform, ultimately driving revenue growth and profitability.
- **Establish Competitive Differentiation:** By offering a recommendation system that surpasses those of competitors we aim to differentiate Stream254 as the go-to destination for personalized entertainment experiences in the Kenyan market
- **Implement Performance Metrics:** We will implement robust metrics and analytics to measure the performance of the recommendation system. This will enable us to continuously optimize algorithms and enhance user engagement over time.

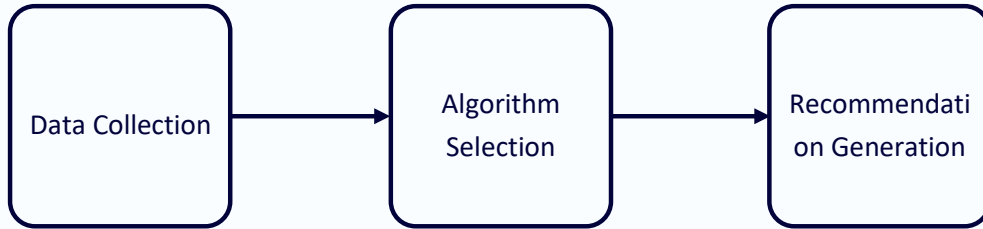
Most Popular Genres



- The most popular genres are as shown: Drama being the most popular and Film Noir being the least popular
- Total number of movies with a mean rating of 4.0 is 1005 .
- Total number of users who have given a rating of 4.0 to these movies is 1350
- The average rating of movies however remains steady over the decades with movies rating averaging between 3 and 4

Behind the Scenes: How it Works

The process of generating recommendations involves three main steps: data collection, algorithm selection, and recommendation generation. **Algorithm Selection** : an appropriate algorithm is selected based on the specific requirements and characteristics of the recommender system. There are various types of algorithms, including



collaborative filtering, content-based filtering, and hybrid approaches

Recommendation Generation: The recommender system generates personalized recommendations for each user.

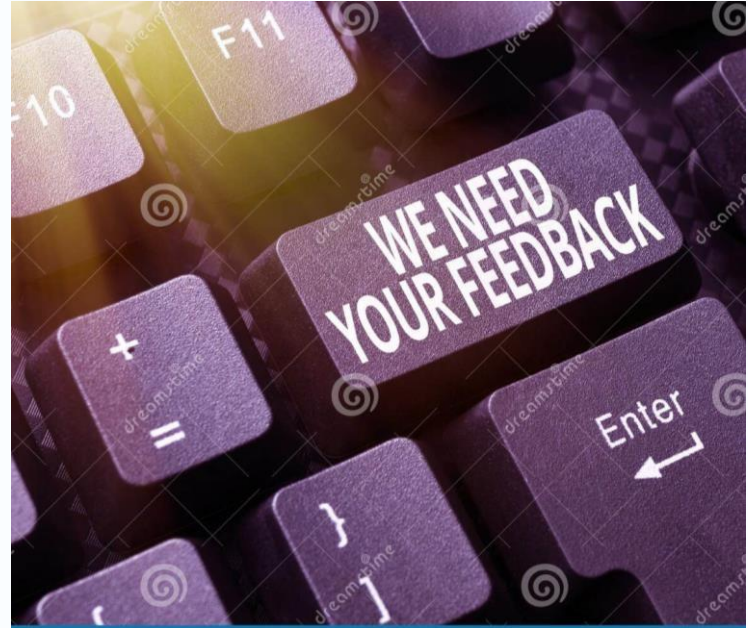
Data Collection: Data is collected from various sources, such as user profiles, browsing history, purchase history, and ratings. This data provides insights into similarities found in the data, and they aim to provide relevant and useful suggestions to the user based on user preferences and behaviors.

The Role of User Feedback

Insight : Understanding user preferences

Refinement : Enhancing recommendations

Engagement : Building loyal user base



Benefits of Our Movie Recommendation System

Personalization: Recommendation systems provide personalized suggestions tailored to individual preferences, enhancing user experience by delivering content that aligns with their interests and needs.

Increased Engagement: By offering relevant recommendations, these systems keep users engaged and encourage them to explore more content, leading to longer browsing sessions and increased interaction with the platform.

Enhanced User Satisfaction: Users appreciate platforms that understand their preferences and make relevant suggestions, leading to higher satisfaction levels and increased loyalty.

Discovery of New Content: Recommendation systems help users discover new movies that they may not have found otherwise, broadening their horizons and introducing them to diverse content.

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Impacts on Movie Industry

Recommender systems have a significant impact on the movie industry, driving increased viewer engagement, facilitating content discovery, enabling targeted marketing efforts, enhancing user experience, generating revenue, informing decision making, and influencing the evolution of content creation practices.

Embrace the magic of personalized movie

recommendations

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Recommendations

- **Understand the Basics:** Recommendation systems analyze data to suggest items like movies based on your past behavior or preferences.
- **Give Feedback:** Rate or like items. Providing feedback helps the system understand your preferences better, leading to more accurate recommendations.
- **Explore Recommendations:** Don't hesitate to explore recommendations! You might discover new things you enjoy based on what the system suggests.

Conclusions

- The system can significantly impact user engagement and platform growth
- Immediate recommendations for new users and items are addressed.
- Provides a pathway for continuous improvement and scalability.
- Well-poised to meet and exceed stakeholder expectations.
- Well-poised to meet and exceed stakeholder expectations.

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